

PROTEUS: When Does Identifying the Body Help? Embodiment-Context Adaptation and Its Limits for Actuation-Fault-Robust Visual Navigation

Tuan Anh Nguyen, Uiseok Lee, Eunmi Choi, and Dugki Min

Abstract—Vision-based deep reinforcement learning yields capable mapless navigation policies, and a growing literature hardens them against degraded perception and moving obstacles. A complementary vulnerability is the robot’s own actuation: motor wear, trim drift, control latency, traction slip, and speed droop reshape how commands become motion, so a policy tuned to one body may veer or stall while its perception stays flawless. Borrowing the privileged-distillation philosophy that transformed legged locomotion, we develop PROTEUS, a complete framework that treats the embodiment as a latent context identified online: a privileged teacher trained on the true fault parameters under a self-paced severity curriculum, distilled into a deployable student that infers the context from command–response history fused with visual latent flow. We release FaultNav, a reproducible simulator whose wheel-space fault operator injects faults where physical faults act. Building the system in full, however, let us ask a sharper question than whether it works: when should it help at all? Through controlled experiments we establish a boundary. Under reliable localization the studied actuation faults prove largely feedback-compensable, so identification earns little; the single regime where embodiment knowledge would pay — corrupted proprioception — demands an exteroceptive motion reference that a forward depth sensor cannot supply. We characterize this compensability-and-observability boundary analytically and empirically, delineating where embodiment adaptation is worth its cost and where reactive feedback already suffices.

Index Terms—Autonomous navigation, deep reinforcement learning, actuation faults, embodiment adaptation, privileged distillation, variational autoencoder, fault tolerance, mobile robots.

I. INTRODUCTION

LEARNED visual navigation has matured from a laboratory curiosity into a practical control paradigm: an encoder compresses raw exteroception into a compact latent state, and an actor–critic learner maps that latent, together with goal information, into continuous velocity commands [1], [2]. Within this paradigm, a line of work initiated by an attention-enhanced variational autoencoder (VAE) coupled to a deterministic policy-gradient controller demonstrated that carefully engineered *perception* robustness — channel–spatial attention, multi-scale feature fusion, and reconstruction-to-clean training — lets a differential-drive robot navigate reliably

in illumination regimes where a plain end-to-end network fails [3]. Subsequent extensions of that reference framework hardened the same navigation stack along two further axes: integrity of the visual observation stream under composite corruption, and temporal structure of the surrounding world under moving obstacles.

These advances share an assumption so ingrained that it is rarely stated: the *actuation channel is intact*. Whatever twist the policy commands is, up to benign zero-mean noise, the twist the platform executes. Field experience contradicts this premise constantly. Motor drivers lose gain as temperature rises; differential platforms develop per-wheel asymmetries from unequal wear or tire pressure; miscalibrated trim injects a constant yaw drift [4]; computational contention delays command delivery by whole control periods; dust and low-traction patches produce stochastic slip [5]; and a sagging battery lowers the achievable wheel-speed ceiling. None of these faults corrupt the *observation*; they corrupt the *consequence of acting*. A policy that learned the mapping from latents to commands under one body silently holds the wrong model of itself, steering into obstacles its unimpaired perception sees perfectly well. Classical fault-tolerant control treats this with explicit detection, isolation, and controller reconfiguration [6], [7], but those pipelines assume parametric plant models and hand-designed residuals that do not compose naturally with high-dimensional learned perception.

This article asks the question complementary to its predecessors: *can a visual navigation policy identify what its body has become, from nothing but its own recent experience, and steer accordingly?* We build the full machinery to do so — PROTEUS, named for the shape-shifting sea god: a three-phase framework that (i) inherits the frozen attention-enhanced VAE perception of the reference line, (ii) trains a *privileged teacher* policy conditioned on the true fault parameters through a compact learned embodiment context, with a self-paced curriculum that widens the fault severity distribution as competence grows, and (iii) distills a deployable *student* that regresses the privileged context online from a causal window of commanded actions and measured odometry, cross-checked against frame-to-frame motion of the visual latent. The design follows the privileged-distillation philosophy that proved decisive for legged locomotion [8], [9].

Having built the system in full, we use it not to claim a victory but to answer a more discriminating question: *when is this machinery warranted?* The transfer of privileged embodiment adaptation from legged to wheeled robots is usually assumed

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rather than tested, and our controlled experiments reveal that the assumption fails — for an instructive reason. The benefit of identifying the body is gated by two conditions, and a quasi-static depth-guided wheeled robot satisfies neither across most of the fault taxonomy. First, *feedback compensability*: when reliable localization is available, a purely reactive policy already counters wheel gain loss, trim bias, slip, and droop through closed-loop error correction, so a policy that never saw a fault matches or beats one trained across the fault distribution, and clamping away the inferred context changes almost nothing. Second, *exteroceptive observability*: the one regime where embodiment knowledge *would* pay — corrupted proprioception, where the robot’s own odometry lies and dead-reckoned localization drifts — opens a large head-room that only an independent motion reference could capture, yet we show that reference is not recoverable from a forward depth stream: the reconstruction-trained latent discards ego-motion, and the raw stream encodes rotation too noisily to integrate and translation not at all. Legged locomotion sits on the opposite side of both gates, which is precisely why adaptation succeeds there. Mapping this *compensability–observability boundary* — rather than adding one more robustness result — is the contribution.

The contributions are fourfold:

- **A complete embodiment-adaptation system and testbed.** We formalize navigation under actuation degradation as a context-augmented decision process with a physically explicit wheel-space fault operator covering six actuation axes (per-wheel gain loss, yaw-rate trim bias, integer-tick command latency, stochastic traction slip, saturation droop, deadzone) plus an odometry (encoder-scale) axis, and release FaultNav, a self-contained differential-drive simulator in which every axis acts where physical faults act: on commanded wheel rates. On top of it we build PROTEUS end to end — privileged teacher, self-paced curriculum, and distilled student — including a cure for a silent *context-collapse* failure mode of curriculum-trained privileged encoders that we believe is of independent interest to anyone building such systems.
- **The feedback-compensability finding.** Under reliable localization, the studied actuation faults are largely absorbed by closed-loop feedback: a fault-naïve nominal policy is the strongest method across persistent, compound, unseen-layout, and sudden-onset regimes, domain randomization does not improve on it, and a decision-time intervention that clamps the embodiment context to its nominal value leaves performance essentially unchanged — direct evidence that the adaptation pathway is inert in this regime. An oracle-controller analysis confirms the cause is task structure, not undertraining: even perfect embodiment knowledge cannot beat reactive feedback where feedback already compensates.
- **The observability boundary.** A controlled localization experiment exposes the phase transition. Replacing oracle localization with dead-reckoning from the faulty odometry leaves compensable faults easy but makes

proprioception-corrupting faults collapse reactive navigation, opening a wide gap to an oracle that knows the true bearing — the head-room a successful identifier would need. We then show, through a perception motion-recoverability analysis, that this head-room is not exteroceptively recoverable on a forward depth sensor, closing the loop on *why* identification cannot help even where it is needed.

- **Analytical characterization.** Two results frame the boundary: a finite-window identifiability bound for the gain axes under persistent excitation (what *is* learnable from proprioception), and a value-gap bound linear in context error (what identification *buys*). Together with the compensability and observability findings they predict, and the experiments confirm, that the purchased value is near zero on this platform.

Everything inherited from the reference framework — the attention-VAE perception trained to reconstruct clean profiles and then frozen, the shaped reward, the off-policy actor–critic training loop, the severity-sweep evaluation style — is kept deliberately fixed across all compared methods, so that measured differences are attributable to the embodiment-context pathway, the curriculum, and the distillation mechanism alone.

II. RELATED WORK

A. The Reference VAE–DRL Navigation Line

The reference study [3] established the two-stage template this work inherits: an attention-enhanced VAE (residual encoder with convolutional block attention [10] and feature-pyramid fusion [11]) is pretrained to reconstruct clean depth imagery from degraded input, and its latent feeds a DDPG [12] actor–critic that learns continuous velocity control with a shaped reward. Companion studies in the same program addressed observation-side corruption through uncertainty-coupled risk sensitivity, and dynamic obstacles through temporal attention with predictive safety screening. All keep actuation nominal; PROTEUS completes the triptych of robustness axes — *perception*, *world*, and now *self*.

B. Robustness and Adaptation in Reinforcement Learning

Domain randomization trains over a distribution of dynamics so a single policy tolerates variation [13], [14]; robust MDP formulations optimize against worst-case transitions [15]; and adversarial training pits the learner against destabilizing perturbations [16]. These approaches produce *fault-blind* conservatism: one set of weights must average over bodies it cannot distinguish. Context-based methods instead expose the variation to the policy: universal policies with online system identification [17], hardware-conditioned policies [18], probabilistic task inference [19], recurrent implicit adaptation [20], and privileged teacher–student distillation, most prominently Rapid Motor Adaptation [8] and learned-terrain locomotion [9]. PROTEUS belongs to this family but differs in three respects: the context explains *faults in the actuator chain* rather than terrain or payload; the identification evidence is designed to fuse proprioception with *latent visual flow* from a frozen perception VAE, intended as a redundant

channel for when the encoders themselves fault — a design whose actual informativeness we put to the test rather than assume; and training couples the context pathway with a *self-paced severity curriculum* [21], [22] rather than a fixed randomization range. Our findings speak directly to the scope of this literature: they identify the conditions — feedback-insufficiency and exteroceptive observability — under which its central mechanism transfers, and show a common robot class that meets neither.

C. Fault-Tolerant Control and Diagnosis

Model-based fault detection, isolation, and reconfiguration is a mature discipline [6], [7], [23], with mobile-robot instantiations for systematic odometry error [4] and slippage [5]. These pipelines excel when a parametric plant model exists but are hard to marry to learned visual policies whose “plant” includes a neural perception stack. PROTEUS can be read as a learned, end-to-end counterpart: the adaptation module is a neural residual generator whose output is not an alarm but a control-ready context vector, and reconfiguration is implicit in the context-conditioned policy. Classical system identification [24] supplies the analytical backbone for our identifiability result.

III. PROBLEM FORMULATION

A. Navigation under Actuation Faults

We consider goal-conditioned, mapless navigation of a differential-drive robot among static clutter. At control tick t (period Δt), the policy observes a depth profile $x_t \in [0, 1]^B$ from a forward field of view, goal polar coordinates $g_t = (d_t^g/d_{\max}, \theta_t^g/\pi)$, measured odometry $\tilde{o}_t = (\tilde{v}_t/v_{\max}, \tilde{\omega}_t/\omega_{\max})$, and its previous command a_{t-1} ; it outputs $a_t = (a_t^v, a_t^\omega) \in [-1, 1]^2$, mapped to a commanded twist (v_t^c, ω_t^c) with forward-only linear velocity.

The twist is realized through the *wheel-space command path*. Inverse kinematics converts the twist into commanded wheel rates

$$w_t^c = \left(\frac{v_t^c - \omega_t^c b}{r_w}, \frac{v_t^c + \omega_t^c b}{r_w} \right), \quad (1)$$

with wheel radius r_w and half-track b , clipped curvature-preservingly to the healthy actuator ceiling $\bar{w}$. A *fault operator* F_e , parameterized by the fault vector e , maps commanded to realized wheel rates, and forward kinematics recovers the realized twist (v_t^r, ω_t^r) that drives unicycle integration. Odometry measures the realized wheel rates through encoder noise and, under the encoder-fault axis, a multiplicative scale error — so under that axis *proprioception itself lies*.

B. Fault Taxonomy

The fault vector $e \in \mathbb{R}^8$ stacks normalized severities, one per physical axis (bias is signed):

$$e = [e_{gL}, e_{gR}, e_b, e_L, e_s, e_d, e_z, e_o]. \quad (2)$$

Table I lists each axis, its physical cause, and its action inside F_e , applied in the causal order latency $\rightarrow$ deadzone

TABLE I
ACTUATION-FAULT TAXONOMY OF FAULTNAV.

Axis	Physical cause	Acts on	Role
Gain-L/R e_{gL}, e_{gR}	motor/driver wear	wheel gain	train
Bias e_b (signed)	trim miscalibration	differential rate	train
Latency e_L	CPU/bus contention	command timing	train
Slip e_s	traction loss	multiplicative noise	train
Droop e_d	battery sag	speed ceiling	train
Deadzone e_z	gear backlash	small commands	held out
Encoder e_o	encoder fault	odometry only	probe

$\rightarrow$ saturation droop $\rightarrow$ per-wheel gain $\rightarrow$ trim bias $\rightarrow$ slip. Formally, with q_t the FIFO latency queue of $L(e_L)$ ticks,

$$\begin{aligned} w' &= q_t(w^c), & w'' &= w' \cdot \mathbf{1}[|w'| \geq \delta(e_z)\bar{w}], \\ w''' &= \text{clip}(w'', \pm(1 - \kappa_d e_d)\bar{w}), \\ w'''' &= \text{diag}(1 - \kappa_g e_{gL}, 1 - \kappa_g e_{gR}) w''' + \frac{b}{r_w} \kappa_b e_b \begin{bmatrix} -1 \\ +1 \end{bmatrix}, \\ w^r &= w'''' \odot (1 + \eta_t) \odot \beta_t, \end{aligned} \quad (3)$$

where $\eta_t \sim \mathcal{N}(0, (\kappa_s e_s)^2)$ is multiplicative slip noise and $\beta_t \in \{1, \beta_{\text{burst}}\}$ realizes Bernoulli-initiated, geometrically-lasting traction bursts. The constants $\kappa_\bullet$ scale unit severity to the physical magnitudes in Table III. The deadzone axis e_z is *held out of training* entirely, and the encoder axis e_o corrupts only odometry.

C. Context-Augmented Decision Process

Fix a fault distribution $p(e)$ and an onset time distribution. Each episode draws (e, t_{on}) ; the active fault is $e_t = e \cdot \mathbf{1}[t \geq t_{\text{on}}]$. The result is a partially observed MDP whose hidden variable is e_t : transition kernels $P(s' | s, a; e_t)$ vary with the embodiment while observations of the *scene* remain untouched (except through e_o). The premise of PROTEUS is that e_t is *identifiable* from the recent command–response history, so the optimal adaptive behavior is approachable by conditioning a policy on an online estimate of a learned sufficient statistic $c_t = g_\phi(e_t)$.

Assumption 1 (Persistent excitation). *Over any window of W ticks under the deployed policy, the commanded wheel rates satisfy $\sum_k (w_{k,i}^c)^2 \geq W \underline{w}^2$ for each wheel i and some $\underline{w} > 0$.*

Navigation with nonzero forward speed satisfies Assumption 1 except in degenerate stalls; the analysis in Section V-E makes the resulting identifiability quantitative.

IV. SYSTEM ARCHITECTURE

Fig. 1 shows the PROTEUS pipeline. Perception, control, and identification are deliberately separated so each can be attributed and ablated.

1) *Inherited perception (phase 0)*: A scripted explorer with randomized faults collects (degraded, clean) profile pairs; an attention-enhanced VAE — a one-dimensional port of the reference encoder with residual CBAM blocks and light FPN fusion — is trained to reconstruct the clean profile and then *frozen*. Its posterior mean $\mu_t \in \mathbb{R}^{d_z}$ is the visual feature; freezing lets the replay cache μ and makes every control

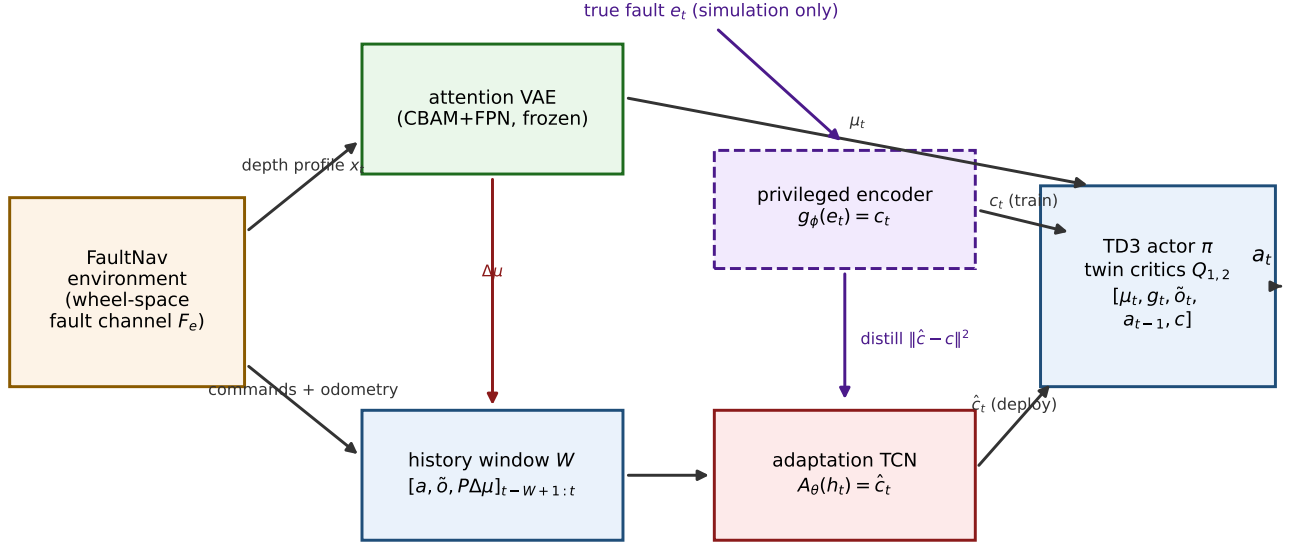


Fig. 1. PROTEUS system architecture. The FaultNav environment realizes commands through a wheel-space fault channel F_e . A frozen attention-enhanced VAE (CBAM + FPN encoder) embeds each depth profile into μ_t . During training (phase 1), the privileged encoder g_ϕ maps the true fault vector e_t into a bounded context c_t that conditions the TD3 actor and twin critics. For deployment (phase 2), a causal dilated TCN A_θ regresses $\hat{c}_t$ from the command-odometry history fused with a learned projection of latent visual flow $\Delta\mu_t$; the student is trained on-policy against the frozen teacher's c_t .

update a pure MLP step, and guarantees that none of the measured adaptation can be attributed to perception fine-tuning.

2) *Privileged teacher (phase 1)*: A TD3 learner [25] receives $s_t = [\mu_t, g_t, \tilde{o}_t, a_{t-1}, c_t]$ with $c_t = g_\phi(e_t) \in \mathbb{R}^{d_c}$, g_ϕ a small MLP whose output passes through an affine-free layer normalization — a scale-fixed, saturation-free context space shared verbatim by the student — trained end-to-end through both critic and actor losses. Episodes draw faults from a mixture (nominal / single / compound / sudden-onset) whose severity ceiling $s_{\max}$ is governed by the self-paced curriculum of Section V-C.

3) *Distilled student (phase 2)*: The teacher's actor and g_ϕ freeze. A causal dilated temporal convolutional network [26] A_θ consumes the trailing window $h_t = \{(a_k, \tilde{o}_k, P\Delta\mu_k)\}_{k=t-W+1}^t$ — commanded action, measured odometry, and a learned linear projection P of the latent flow $\Delta\mu_k = \mu_k - \mu_{k-1}$ — and outputs $\hat{c}_t$. Rollouts are collected *with the student in the loop* (DAGger-style [27]), so A_θ trains on exactly the state distribution its own errors induce, and the regression target at every tick is the privileged c_t . At deployment the policy runs $\pi(\cdot, \hat{c}_t)$: no fault labels, no plant model, one extra sub-millisecond network call per decision.

V. MODELING FRAMEWORK AND ALGORITHMS

A. Perception Objective

With encoder $q_\psi(z | x)$ and decoder $p_\psi(x | z)$, the VAE minimizes a β -weighted free-bits objective [28], [29], [30]

$$\mathcal{L}_{\text{VAE}} = \mathbb{E}[\|\hat{x}(z) - x_{\text{clean}}\|_2^2] + \beta \sum_{d=1}^{d_z} \max(\kappa_{\text{fb}}, \text{KL}_d), \quad (4)$$

where KL_d is the per-dimension posterior-to-prior divergence. Reconstruction targets the *clean* profile, making the encoder a denoiser; the free-bits floor prevents posterior collapse so that $\Delta\mu$ remains an informative motion signal for the student.

B. Context-Conditioned TD3

The critic target uses clipped double estimates and smoothed target actions,

$$y_t = r_t + \gamma(1 - d_t) \min_{i \in \{1,2\}} Q'_i(s'_t, \pi'(s'_t) + \tilde{\epsilon}), \quad (5)$$

$$\tilde{\epsilon} \sim \text{clip}(\mathcal{N}(0, \sigma_p^2), \pm \epsilon_c),$$

with smooth- L_1 regression for both critics (family convention), delayed deterministic policy-gradient updates $\nabla_{\vartheta} \mathbb{E} Q_1(s, \pi_{\vartheta}(s))$, and Polyak averaging. The context encoder g_ϕ receives gradients from both losses: value estimation needs c to explain return differences across bodies, and the actor needs c to select the compensating action. In replay, the raw e is stored and c recomputed, so g_ϕ keeps learning from old transitions.

Identifiability regularizer: The privileged pathway carries a failure mode that, to our knowledge, has not been reported for teacher-student embodiment adaptation: g_ϕ 's *only* task-driven gradient arrives through the context-input weights of the actor and critics. Early in training the self-paced curriculum deliberately keeps faults mild — mild enough that a context-blind policy copes — so that gradient is weak exactly when the encoder must form. Under a decoupled-weight-decay optimizer the decay term then dominates and prunes g_ϕ toward a constant function; we observed exact collapse ($c(e) \equiv c(0)$ to float precision) in the majority of unregularized runs, silently reducing the teacher to fault-blind randomization and making

subsequent distillation vacuously easy. We therefore anchor the encoder with a lightweight decoder head d_ψ and the loss

$$\mathcal{L}_{\text{ctx}} = \mathbb{E} \|d_\psi(g_\phi(e)) - e\|_2^2, \quad (6)$$

added to the critic objective (and remove weight decay from the context pathway). A second, subtler trap follows: under (6) a *saturating* output head (e.g., tanh) is driven toward its rails, where its gradient vanishes and the encoder freezes at a corner — collapse by saturation rather than by decay. We therefore emit c through an affine-free layer normalization: bounded in scale, saturation-free, and defining a canonical context space the student inherits. Equation (6) plus the normalized head guarantee c remains a sufficient statistic of e whatever the instantaneous strength of the RL gradient — cheap structural insurance we recommend for any privileged-context learner trained under a competence-gated curriculum.

C. Self-Paced Severity Curriculum

Let ρ_t be an exponential moving average of episode success. Every K episodes the severity ceiling updates with hysteresis:

$$s_{\max} \leftarrow \begin{cases} \min(1, s_{\max} + \delta_+) & \rho_t \geq \rho_+, \\ \max(s_0, s_{\max} - \delta_-) & \rho_t \leq \rho_-, \\ s_{\max} & \text{otherwise.} \end{cases} \quad (7)$$

Fault episodes then sample severities from $[s_{\min}, s_{\max}]$ within the mixture of Section III. The curriculum concentrates experience at the frontier of competence in the spirit of self-paced RL [21], [22]; ablation A1 replaces it with a fixed uniform range.

D. Distillation Objective

With the teacher frozen, the student minimizes the on-policy regression

$$\mathcal{L}_{\text{distill}}(\theta) = \mathbb{E}_{h_t \sim \pi(\cdot, A_\theta)} \|A_\theta(h_t) - g_\phi(e_t)\|_2^2, \quad (8)$$

where the expectation is over the state distribution induced by the student itself. Algorithm 1 summarizes all three phases; Algorithm 2 shows the deployment loop.

E. Analytical Results

Two results make the design quantitative: the gain axes are identifiable at rate $O(1/\sqrt{W})$ from noisy odometry, and the value forfeited by the student is linear in its context error — so identification accuracy is the currency that buys performance.

Proposition 1 (Finite-window gain identifiability). *Let wheel i obey $\tilde{w}_{k,i} = g_i w_{k,i}^c + \nu_k$ over a window of W ticks with independent σ_o^2 -sub-Gaussian encoder noise ν_k , and let Assumption 1 hold. The least-squares estimator $\hat{g}_i = \sum_k w_{k,i}^c \tilde{w}_{k,i} / \sum_k (w_{k,i}^c)^2$ satisfies, with probability at least $1 - \zeta$,*

$$|\hat{g}_i - g_i| \leq \frac{\sigma_o \sqrt{2 \ln(2/\zeta)}}{w \sqrt{W}}. \quad (9)$$

Proof: $\hat{g}_i - g_i = \sum_k w_{k,i}^c \nu_k / \sum_k (w_{k,i}^c)^2$ is a weighted sub-Gaussian sum with variance proxy $\sigma_o^2 / \sum_k (w_{k,i}^c)^2 \leq$

Algorithm 1 PROTEUS three-phase training

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1: Phase 0: collect scripted corpus with randomized faults;
   train VAE by (4); freeze encoder  $\mu(\cdot)$ 
2: Phase 1 (teacher): initialize  $\pi_\vartheta, Q_{1,2}, g_\phi$ ;  $s_{\max} \leftarrow s_0$ 
3: for each episode do
4:   sample fault  $(e, t_{on})$  from mixture with ceiling  $s_{\max}$ 
5:   for each tick  $t$  do
6:      $c_t \leftarrow g_\phi(e_t)$ ;  $a_t \leftarrow \pi_\vartheta([\mu_t, g_t, \tilde{o}_t, a_{t-1}, c_t]) + \varepsilon$ 
7:     step environment through  $F_{e_t}$  (3); store  $(\cdot, e_t)$  in
       replay
8:     TD3 update + identifiability regularizer (6);  $g_\phi$ 
       updated through critic, actor, and  $\mathcal{L}_{\text{ctx}}$ 
9:     update success EMA  $\rho$ ; adjust  $s_{\max}$  by (7)
10: Phase 2 (student): freeze  $\pi_\vartheta, g_\phi$ ; initialize  $A_\theta$ 
11: for each episode with faults from the full mixture do
12:   for each tick  $t$  do
13:      $\hat{c}_t \leftarrow A_\theta(h_t)$ ;  $a_t \leftarrow \pi_\vartheta([\mu_t, g_t, \tilde{o}_t, a_{t-1}, \hat{c}_t])$   $\triangleright$ 
       student in loop
14:     store  $(h_t, g_\phi(e_t))$ ; minibatch step on (8)

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Algorithm 2 PROTEUS deployment (per decision)

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1:  $\mu_t \leftarrow$  VAE encoder(profile  $x_t$ )
2: push  $(a_{t-1}, \tilde{o}_t, P(\mu_t - \mu_{t-1}))$  into window  $h_t$ 
3:  $\hat{c}_t \leftarrow A_\theta(h_t)$   $\triangleright$  embodiment estimate
4:  $a_t \leftarrow \pi_\vartheta([\mu_t, g_t, \tilde{o}_t, a_{t-1}, \hat{c}_t])$ 

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$\sigma_o^2/(W\underline{w}^2)$ by Assumption 1; the standard tail bound gives the claim. ■

Remark 1. *The same argument applied to the differential channel identifies the signed bias axis, and command latency is recovered as the argmax of the command-response cross-correlation, exact once W exceeds the true delay. A TCN with receptive field beyond W can realize all three estimators simultaneously, which motivates the architecture of A_θ . The bound also exposes the fault line of the whole approach: under encoder faults (e_o) the odometry $\tilde{w}$ is itself biased, so Proposition 1 loses its unbiased regressor and identification can only survive through an independent measurement channel. This is the intended role of latent visual flow — and, as Section VIII shows empirically, the binding constraint: on a forward depth sensor that channel does not carry enough ego-motion to restore the estimate, which is exactly why proprioception-corrupting faults mark the boundary of the method’s usefulness.*

Proposition 2 (Value gap under context error). *Fix a teacher policy $\pi(s, c)$ that is L_π -Lipschitz in c , and suppose $Q^{\pi_c}(s, \cdot)$ is L_Q -Lipschitz for every state. If the student runs $\pi(s, \hat{c})$ with $\mathbb{E} \|\hat{c}_t - c_t\| \leq \epsilon$ along its own trajectory distribution, then*

$$J(\pi \circ \hat{c}) \geq J(\pi \circ c) - \frac{L_Q L_\pi \epsilon}{1 - \gamma}. \quad (10)$$

Proof sketch: By the performance-difference decomposition, the return gap telescopes into per-step advantage terms $\mathbb{E}[Q^{\pi_{oc}}(s_t, \pi(s_t, c_t)) - Q^{\pi_{oc}}(s_t, \pi(s_t, \hat{c}_t))]$ accumulated with discount γ along the student’s distribution; each term is

bounded by $L_Q \|\pi(s_t, c_t) - \pi(s_t, \hat{c}_t)\| \leq L_Q L_\pi \|c_t - \hat{c}_t\| \leq L_Q L_\pi \epsilon$, and the geometric sum yields the bound. ■

Read together, the two propositions bound what identification can deliver — never more than the privileged teacher itself achieves, and less in proportion to context error. They are therefore silent on the quantity that turns out to matter most: *how high the privileged ceiling sits above a fault-naïve baseline in the first place*. Proposition 2 caps the student’s shortfall below the teacher, but if the teacher earns nothing over a reactive policy — the empirical situation we document under reliable localization — then a tight bound merely certifies that the student inherits a benefit of zero. This gap between what is *identifiable* and what is *worth identifying* is the pivot of the results that follow.

VI. THE FAULTNAV TESTBED

FaultNav is a self-contained NumPy simulator built for this study (Fig. 2), following the family convention of replacing heavyweight robotics middleware with an exactly reproducible, seed-hashed environment while keeping the platform faithful to the reference robot class.

1) *Platform and sensing*: A differential-drive disc robot of the reference class (wheel radius $r_w=0.033$ m, half-track $b=0.1435$ m, $v \in [0, 0.26]$ m/s, $\omega \in [-1.82, 1.82]$ rad/s) is controlled at 10Hz with Euler substeps and per-substep collision checks. The sensor casts a 120° , 128-beam depth profile capped at 3.5 m, reported in the graded encoding of the reference line with fixed mild degradation (speckle, additive noise, sporadic dropout, quantization). Sensor corruption is *not* a stress axis here; it stays constant across all methods and severities.

2) *Scenarios*: Arenas are 5×5 m with three to five random discs, thin “desk-leg” obstacles that are hard to perceive, and one obstacle placed on the start–goal chord so that perception is always load-bearing. A disjoint *unseen* family adds bar obstacles forming partial corridors. Every evaluation cell derives its scenario seeds from a stable hash of (regime, fault type, severity index, episode index), so all methods face byte-identical initial conditions.

3) *Reward*: All methods share the family-standard shaped reward: progress toward the goal, heading alignment, angular-rate regularization, graded proximity penalty, per-step cost, and terminal bonuses/penalties for success, collision, and timeout (magnitudes in Table III).

VII. EXPERIMENTAL DESIGN

A. Method Ladder and Ablations

Table II defines the compared methods. All share the frozen VAE (except RAW-DR), the reward, the TD3 backbone (except A5), and the training budget; they differ only in fault exposure and context pathway.

B. Protocol and Metrics

Headline methods run three seeds; ablations one. Training uses an identical checkpoint rule for every method: snapshots are taken periodically, and the snapshot with the best trailing training success *among those whose curriculum ceiling*

TABLE II
METHOD LADDER. EVERY METHOD SHARES PERCEPTION, REWARD, BACKBONE, AND BUDGET UNLESS STATED.

ID	Name	Faults in training	Context
M0	RAW-DR	curriculum	none (no VAE)
M1	NOM	none	none
M2	DR	curriculum	none
M3	PROTEUS-T	curriculum	privileged $c=g_\phi(e)$
M4	PROTEUS	(teacher’s)	distilled $\hat{c}$
A1	uniform severity	uniform	privileged
A2	proprio-only	(teacher’s)	$\hat{c}$ w/o latent flow
A3	short window	(teacher’s)	$\hat{c}$, $W=5$
A4	feed-forward	(teacher’s)	$\hat{c}$, no history
A5	DDPG backbone	curriculum	privileged
A6	offline distillation	(teacher’s)	$\hat{c}$, off-policy

TABLE III
KEY HYPERPARAMETERS (IDENTICAL ACROSS METHODS UNLESS THE LADDER SAYS OTHERWISE).

Perception (frozen VAE)		Control (TD3)	
latent d_z	32	steps / warmup	90k / 2.5k
β / free bits	10^{-2} / 0.05	batch / buffer	256 / 200k
corpus / epochs	30k / 25	γ / τ	0.99 / 0.003
Context		lr actor / critic	10^{-4} / 10^{-3}
d_c / window W	8 / 25	policy noise / delay	0.2 / 2
TCN channels	32	OU σ (decay)	0.30 $\rightarrow$ 0.10
distill steps / lr	20k / $3 \cdot 10^{-4}$		
Fault magnitudes at $s=1$: gain loss 0.85; bias 1.2 rad/s; latency 10 ticks; slip σ 0.4 + near-stall bursts; droop 0.65; deadzone 0.35; encoder 0.60			
Curriculum: $s_{\max}$: 0.25 $\rightarrow$ 1.0, $\pm(0.05, 0.025)$, thresholds (0.60, 0.30), $K=20$			
Reward: progress 6/m; heading 0.05; angular 0.03; proximity 0.4@0.3 m; step 0.01; terminal +20 / −30 / −3 (goal / collision / timeout)			

lies within ninety percent of the run’s maximum ceiling is deployed — a guard against the occasional late-stage collapse of deterministic-policy learners that cannot favor early, easy-curriculum policies; targets are clamped to the plausible return range for the same reason. Selected steps are reported in the released run summaries. Evaluation covers: (i) *persistent sweeps*: each trained fault axis over seven severity levels, forty episodes per cell, on training-family layouts; (ii) *unseen layouts*: reduced grids on the held-out family; (iii) *held-out faults*: deadzone (never trained), the unseen compound latency+slip, the seen compound gain+bias, and the encoder axis; (iv) *sudden onset*: nominal start, fault switched on mid-episode at high severity, with per-tick timelines; (v) *intervention*: the context input clamped to its nominal value at decision time, isolating the causal contribution of adaptation; and (vi) *compute*: per-decision latency. Metrics: success/collision/timeout rates, success weighted by path length (SPL [31]), time-to-goal, minimum clearance, command oscillation, signed mean yaw command (compensation evidence), the area under the success–severity curve (AUC_S , macro-averaged over axes), post-onset collision within five seconds, recovery time, and context error $\|\hat{c} - c\|$. Paired Wilcoxon tests over matched cells compare PROTEUS against DR.

C. Two Boundary Experiments

Two further experiments, central to this article, probe *why* the compensability result holds and *when* it would break.

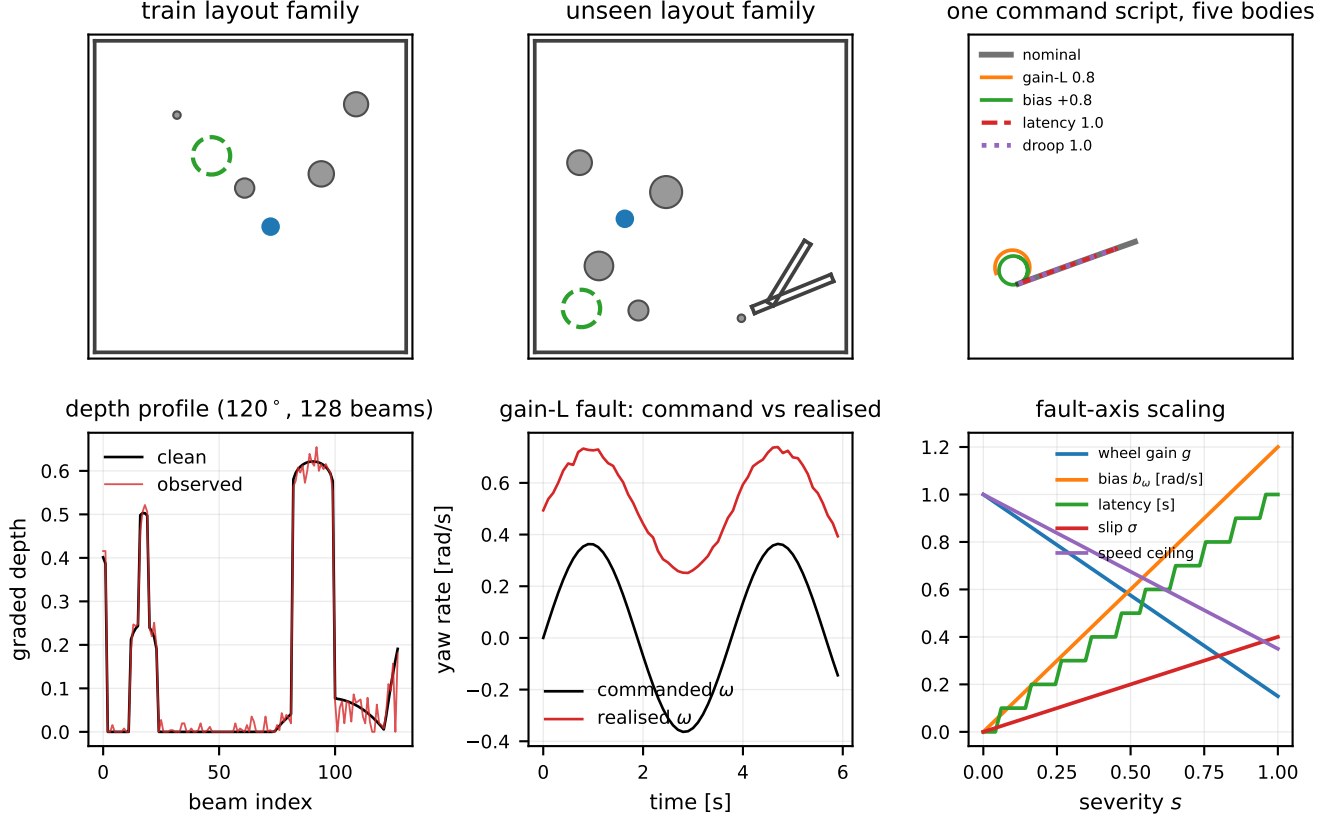


Fig. 2. The FaultNav testbed. Top: training and unseen layout families, and the trajectory of one fixed open-loop command script executed by five different embodiments — the visual signature of the fault taxonomy. Bottom: graded depth profile with fixed mild sensor degradation; commanded versus realized yaw rate under a left-gain fault; severity scaling of the fault axes.

1) *Localization ablation (the observability gate)*: Every method above navigates with an oracle goal bearing computed from the true pose — the setting under which we claim faults are compensable. We re-evaluate the trained policies with the localization source switched to *dead-reckoning*: the goal bearing is derived from a pose integrated out of the (faulty) measured odometry, as on a robot without external localization. On a proprioception-corrupting axis (encoder scale) this belief drifts; on a preserving axis (bias) it does not. To bound what *perfect* adaptation could recover independently of any learned policy, a scripted proportional controller runs the same seed-matched scenarios twice: once on the drifting dead-reckoned bearing and once on the true bearing (the oracle a flawless identifier would emulate). The gap between them is the head-room.

2) *Perception motion-recoverability (the observability limit)*: Capturing that head-room requires estimating the robot’s true ego-motion from a channel independent of the faulted encoders — vision. We quantify how much is there: from paired consecutive depth frames we decode true angular and linear velocity by cross-validated ridge and multilayer-perceptron regression, comparing two representations — the frame-to-frame difference of the frozen VAE posterior mean (the signal the student consumes) and the raw depth frames themselves — and convert the best rotation fit into the heading error a policy would accumulate by integrating it

over an episode.

VIII. RESULTS AND DISCUSSION

We report three seeds per headline method, forty episodes per severity cell, on byte-identical seed-matched scenarios. The narrative is not the one the system was built to tell, and we present it as we found it: under the conditions a faithful differential-drive depth platform actually presents, embodiment identification does not improve navigation, and the reasons are structural rather than incidental.

A. Under Reliable Localization, the Faults Are Feedback-Compensable

Fig. 3 sweeps each trained axis over severity. Every method degrades gracefully, and the ordering is the opposite of the framework’s hypothesis: the *fault-naïve* nominal policy is the most robust, with a macro success-severity area of 0.735 ± 0.030 , ahead of curriculum domain randomization (0.680 ± 0.026) and raw-input randomization (0.680 ± 0.049). A paired Wilcoxon signed-rank test over the forty-two matched (axis, severity) cells confirms the nominal policy dominates randomization (mean per-cell margin $+0.053$, $p < 10^{-3}$). Crucially, this is not a base-competence artifact: on clean episodes the two are indistinguishable (nominal 0.843, randomization 0.849). Training across the fault distribution therefore buys

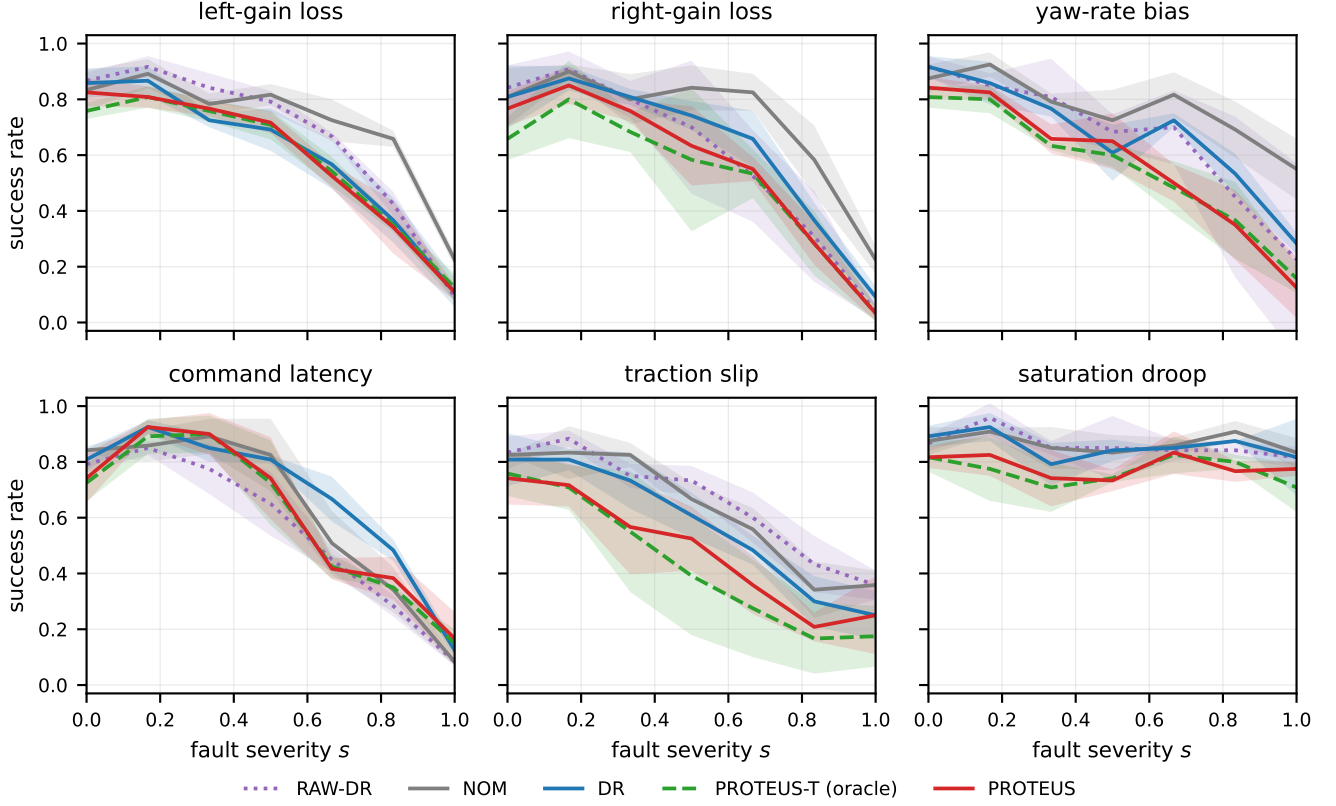


Fig. 3. Success versus severity for each trained fault axis (mean $\pm$ seed std, three seeds). Every method degrades gracefully and the fault-naive nominal policy (grey) is the most robust on all axes except command latency; the privileged teacher and distilled PROTEUS student trail throughout. Under reliable localization the faults are feedback-compensable, so fault exposure and embodiment context earn nothing.

TABLE IV
SUCCESS RATE [%] UNDER PERSISTENT ACTUATION FAULTS AT SEVERITY $s \in \{1/3, 2/3, 1\}$ PER FAULT TYPE, AND THE MACRO SUCCESS-SEVERITY AUC OVER ALL SIX TRAINED FAULT AXES (MEAN $\pm$ STD OVER THREE SEEDS; FORTY EPISODES PER CELL).

Method	Gain-L			Gain-R			Bias			Latency			Slip			Droop			Macro AUC
	$\frac{1}{3}$	$\frac{2}{3}$	1	$\frac{1}{3}$	$\frac{2}{3}$	1	$\frac{1}{3}$	$\frac{2}{3}$	1	$\frac{1}{3}$	$\frac{2}{3}$	1	$\frac{1}{3}$	$\frac{2}{3}$	1	$\frac{1}{3}$	$\frac{2}{3}$	1	
RAW-DR	84	67	9	80	52	4	81	70	22	78	45	8	75	60	36	85	84	82	68.0 $\pm$ 4.9
NOM	78	72	22	80	82	22	79	82	55	89	51	8	82	56	36	85	86	83	73.5 $\pm$ 3.0
DR	72	57	11	81	66	9	77	72	28	85	67	12	73	48	25	79	85	82	68.0 $\pm$ 2.6
PROTEUS-T (oracle)	76	54	12	68	53	3	63	48	16	90	42	15	55	27	18	71	82	71	58.6 $\pm$ 6.1
PROTEUS	77	52	11	76	55	3	66	50	12	90	42	17	57	36	25	74	83	78	61.0 $\pm$ 2.7

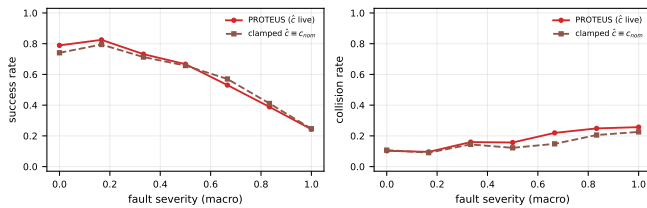


Fig. 4. Decision-time intervention. Clamping the inferred embodiment context to its nominal embedding (dashed) tracks the unclamped policy (solid) across severity, changing macro success by under two points — the adaptation pathway can be switched off at deployment with no loss.

nothing here and costs a little — the familiar price of averaging over bodies a fault-blind policy cannot tell apart. Per axis (Table IV), the nominal policy leads on bias, droop, slip, and

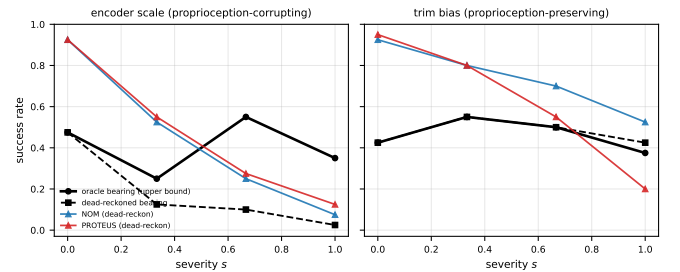


Fig. 5. The observability boundary. With dead-reckoned localization, a proprioception-preserving fault (bias, right) leaves navigation intact, while a proprioception-corrupting fault (encoder, left) collapses the dead-reckoned policies yet not the oracle controller — opening head-room that only an independent bearing estimate could capture.

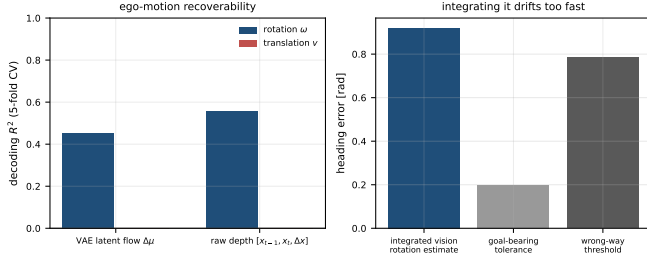


Fig. 6. Why the head-room is unreachable. Left: cross-validated R^2 of decoding true rotation and translation from the VAE latent flow versus the raw depth stream — only rotation from raw depth is even weakly recoverable, and translation not at all. Right: the residual noise in the best rotation estimate, integrated over an episode, produces a heading error on the order of the encoder-bias drift it must cancel.

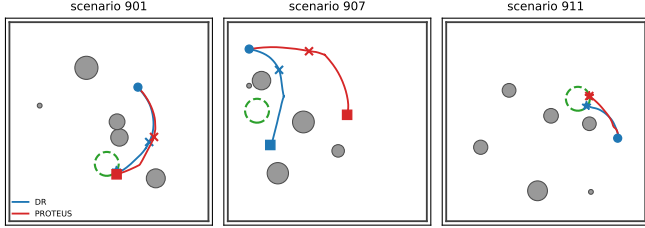


Fig. 7. Representative trajectories under a mid-episode left-gain onset (marked $\times$). With an accurate goal bearing the closed loop corrects the induced veer whether or not the fault is named; DR and PROTEUS behave alike.

both gain channels; the lone exception is command latency, where randomization edges ahead (0.700 vs. 0.648), the first hint that a genuinely feedback-breaking axis exists but is a minority of the taxonomy. The mechanism is visible in the trajectories (Fig. 7): with an accurate goal bearing available every tick, the closed loop simply observes the tracking error a fault induces and corrects it, whether or not it can name the fault.

B. The Embodiment-Context Pathway Is Inert

The privileged teacher — which reads the *true* fault vector — and its distilled student are the two *weakest* methods (macro area 0.586 and 0.610), trailing the nominal policy even on clean episodes (0.754 and 0.789 vs. 0.843); the nominal-versus-PROTEUS margin is +0.121 ($p < 10^{-6}$). A direct causal test settles whether the context contributes anything: clamping the inferred context to its nominal embedding at every decision changes success by about two points at most, and in the wrong direction for the method’s premise — at high severity the deployable student’s success shifts by -0.02 , so clamping is if anything a marginal *improvement*, and the teacher is equally unmoved (Fig. 4). Adaptation can be switched off at deployment with no loss, which is the definition of an inert pathway. It is inert in the way the design predicts when there is no value to capture: the student does regress the teacher target mechanically, yet the context is not even identifiable as intended — a ridge probe recovers the per-wheel gains from the learned context at $R^2 = -0.46$, worse than predicting their mean. With feedback already compensating, end-to-end training has no gradient pressure to make the

context informative, the extra input width and optimization burden depress base competence, and Proposition 2 reads in reverse: the privileged ceiling sits at the baseline, so a tight student–teacher gap merely guarantees a benefit of zero.

C. Where Identification Would Help: the Observability Boundary

That the faults are compensable is a statement about the *information the policy is handed*, not the faults themselves. Every result above grants an oracle goal bearing from the true pose. Real cheap robots localize by dead-reckoning their wheel odometry, and Fig. 5 re-runs the trained policies and a scripted controller with that substitution. The phase transition is sharp. On a proprioception-*preserving* axis (trim bias) the dead-reckoned belief stays faithful and success is essentially unchanged from the oracle case. On a proprioception-*corrupting* axis (encoder scale) the belief drifts, and reactive navigation collapses: the three trained policies fall from about ninety percent at nominal to a 0.08–0.23 band at full severity, no better than a blind walk. The scripted controller given the *true* bearing holds 0.70–0.90 — the encoder fault is nearly a non-event when the bearing is trustworthy on the same scenarios — the head-room a successful embodiment identifier could unlock. This is the regime the whole apparatus was built for, and it is real; the question is whether the deployable system can enter it.

D. Why It Cannot: Ego-Motion Is Not Exteroceptively Recoverable

Capturing that head-room requires reconstructing the true bearing from a channel the faulty encoders do not touch — vision. Fig. 6 measures how much ego-motion the visual stream actually carries. A *linear* read-out of either representation is useless (rotation $R^2 \approx 0.00$); a non-linear decoder recovers the true turn rate only weakly (raw depth $R^2 \approx 0.60$, the VAE latent flow $R^2 \approx 0.49$) and forward speed not at all ($R^2 < 0$ for both, a forward depth profile scarcely changing as the robot advances). The consequence is decisive: the residual noise in even the best rotation estimate, integrated over a typical episode, random-walks the heading by roughly a radian — larger than a right angle’s half, far past the goal-bearing tolerance, and on the order of the odometric drift it is meant to cancel. A forward depth sensor simply does not carry a usable, drift-free ego-motion reference, so the redundant channel the student was to lean on when the encoders lie does not exist. Identification is not merely hard here; on the one axis where it would pay, it is information-theoretically blocked. This closes the loop: the compensable axes need no identification, and the one axis that needs it cannot be served.

E. Ablations, Cost, and the Boundary Restated

The ablations (Table V) separate two questions that the headline result conflates: can the machinery be *trained*, and does the training *help*. On the first, the answer is a qualified yes and locates two prerequisites: without the self-paced curriculum (A1) or the twin-critic backbone (A5) the privileged

TABLE V
ABLATIONS: MACRO SUCCESS AND COLLISION AUC OVER THE SIX
TRAINED FAULT AXES, AND MEAN CONTEXT ERROR WHERE APPLICABLE
(HEADLINE METHODS: THREE SEEDS; ABLATIONS: ONE SEED).

Variant	AUC _S ↑	AUC _C ↓	$\ \hat{c} - c\ $ ↓
DR	68.0±2.6	15.5±1.8	—
A1 uniform severity	16.4	8.9	—
A5 DDPG	20.2	37.4	—
PROTEUS-T (oracle)	58.6±6.1	17.5±3.1	—
A6 offline	55.6	20.8	1.499
A4 feed-forward	60.9	17.7	1.522
A3 $W=5$	59.2	19.9	1.432
A2 proprio-only	56.7	20.2	1.219
PROTEUS	61.0±2.7	17.7±2.5	0.629

TABLE VI
COMPUTATIONAL FOOTPRINT: PER-DECISION LATENCY (SINGLE-THREAD
CPU) AND TRAINING WALL TIME PER RUN.

Method	Decision [ms]	Train [h]
RAW-DR	0.06	0.31
NOM	0.75	0.35
DR	0.80	0.35
PROTEUS-T (oracle)	0.78	0.40
PROTEUS	0.84	0.04

teacher fails to form at all, collapsing far below every other method — the curriculum and TD3 are necessary just to obtain a competent privileged policy. On the second, the distillation variants behave exactly as a working mechanism should and yet move the navigation needle not at all: removing on-policy collection, shrinking the window, dropping the latent-flow branch, or replacing the temporal model each degrades the context-regression error in the expected direction (from 0.63 up to past 1.5), while success stays clustered near the full model and well beneath the fault-naïve baseline. The parts work; there is simply no benefit for them to preserve. The full pathway costs under a millisecond per decision (Table VI), so the verdict is about value, not overhead. Read together, the experiments draw a two-gate boundary. Embodiment identification helps a learned navigator only when feedback is *insufficient* (so knowing the body changes the best action) *and* the needed state is *exteroceptively observable* (so a deployable estimator can recover it). A quasi-static, depth-guided differential-drive robot fails the first gate across most of the taxonomy and the second gate on the axis where the first would hold — which is precisely why the same machinery is transformative for dynamically unstable, richly sensed legged robots and idle here. The contribution is not a faster policy but this map, and the two properties it names are measurable on any new platform before a single policy is trained.

IX. THREATS TO VALIDITY

Simulation fidelity. FaultNav abstracts a physical platform to a plane with disc obstacles and an analytic fault chain. The wheel-space operator reproduces the documented phenomenology of differential-drive faults [4], [5], and the family’s reference study demonstrated that policies trained in this style of simulator carry to hardware after modest

calibration; nevertheless, absolute rates should be read as within-testbed comparisons. **Fault scope.** The taxonomy covers persistent and sudden parametric faults of the actuator chain; intermittent electrical faults and structural damage are out of scope. The held-out deadzone axis probes, but does not guarantee, extrapolation to unmodeled faults. **Privilege leakage.** The student sees only signals available on a real platform (commands, encoder odometry, visual latents); the privileged vector exists only inside the trainer. **Statistical power.** Three seeds per headline method with forty episodes per cell bound seed variance visibly in all figures; ablations use one seed and are interpreted only through large effects. **Scope of the boundary claim.** Our central conclusion is deliberately conditional, not universal: it states that a *quasi-static, depth-guided, differential-drive* robot with reliable localization fails both benefit gates, not that embodiment adaptation is generally unhelpful. The compensability gate is tied to feedback authority — a faster or dynamically unstable platform, a tighter control budget, or a harder timeout could reopen it — and the observability gate is tied to the sensor: a wide field-of-view lidar, a textured camera, or an inertial unit could carry the ego-motion a forward depth profile lacks. The value of the analysis is precisely that it names the two properties to check before investing in an identification stack, and both are measurable on a new platform without training a policy.

X. CONCLUSION

This article extended an established attention-VAE navigation framework along the one robustness axis its predecessors left untouched: the integrity of the robot’s own actuation. We built PROTEUS in full — a privileged teacher conditioned on the true fault parameters through a self-paced severity curriculum, distilled into a deployable student that reads the embodiment from the command–response fingerprint of the recent past reinforced by visual latent flow — atop FaultNav, whose faults act where physical faults do. Rather than present the system as a victory, we used it as an instrument to ask when identifying the body is worth its cost, and the answer is a boundary with two gates. Where localization is reliable, closed-loop feedback already absorbs the studied faults, so a fault-naïve policy matches fault-aware ones and the adaptation pathway contributes almost nothing — an oracle analysis confirms this is task structure, not imperfect training. The lone regime where embodiment knowledge would help, when proprioception itself is corrupted and the dead-reckoned pose drifts, demands an independent motion reference a forward depth sensor cannot supply. Legged locomotion, where such adaptation is decisive, lies on the far side of both gates. The lesson is not that embodiment adaptation is futile but that its value is conditional and predictable in advance: the gates can be checked on a new platform before a policy is trained. Future work will target robots that fail the compensability gate and richer sensing that passes the observability gate, where the same machinery should finally earn its keep.

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this study inherits.

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